HARDBOUND PRINTING RESERVATION & MONITORING SYSTEM FOR FOUR ANGEL'S COPY CENTER

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ABSTRACT

This project proposes a software based Hardbound Printing Reservation & Monitoring System for Four Angel's Copy Center. For this project, the printing center is assumed to handle student printing reservations and claiming schedules using manual processes. The proposed system is intended to organize and monitor reservation requests, printing schedules, claiming schedules, and reservation details between students and printing staff. The system is intended to use Java GUI for the user interface and core reservation functions, NetBeans IDE for developing and running the appliction, MySQL for data storage, and phpMyAdmin for database administration during development and maintenance..

The project will use an Agile SDLC approach supported by Scrum practices. Development will be divided into small stages so that the team can present working features, collect feedback, and improve the system before final deployment. The project documentation also considers requirements, feasibility, possible risks, project schedule, estimated costs, and Scrum activities.

I. PROJECT BACKGROUND

1.1 Client / Organization Background

For this project, Four Angel's Copy Center is a printing service that provides hardbound printing services to students. The main users of the proposed system are the students who submit printing reservations and the printing staff who manage the requests. The system is planned as a software application that can be installed on the printing center's computer.

1.2 Current Process

The current process starts when a student goes to the printing center and submits their file or hardbound printing request to the staff. The staff receives the file, asks for the necessary details, and writes or records the information. The staff then processes the printing request based on the information provided while the student waits for the printing to be completed. At the end of the transaction, the payment and order details may be recorded separately for monitoring.

1.3 Identified Problems

Based on the assumed process, the following problems may occur:

- Handwritten printing request may be difficult to read or may contain incomplete or inaccurate details.
- Manual processing of printing requests can increase the possibility of delays and errors.
- Printing information may be delayed or misunderstood when passed verbally.
- It may be difficult to review previous printing requests when records are not stored in one system.
- Preparing and monitoring printing and claiming schedules can take additional time when records are manual.
- Long queues may occur when several students submit or claim printing requests at the same time.

2. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of the project is to create a simple software based Printing Reservation and Monitoring System that can help students submit printing requests, monitor reservation statuses, organize claiming schedules, and allow printing staff to manage reservations efficiently while reducing manual work and minimizing queues.

2.2 Main Project Objective

To design and develop a software based Hardbound Printing Reservation & Monitoring System for Four Angel's Copy Center that supports faster, more organized, and easier to manage printing reservation processing.

2.3 Specific Objectives

· To allow students to create an account and log in securely.

· To allow students to enter file and printing details through a digital reservation form

· To allow students to select printing and claiming schedules and view their reservation status.

· To allow authorized printing staff (admin) to view submitted reservations and approve or decline requests.

· To store reservation information for easier record management.

· To reduce delays, long queues, and miscommunication caused by manual reservation processing.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

The proposed desktop application will include the following major functions:

User / Area	Included Functions
Student	Register and log in, enter file details, select service type, binding, paper size, color, paper type, binding, document type, quantity, claiming data and optional time, add notes, submit a reservation, view reservation status.
Printing Staff/Admin	Secure login, view costumer list, view submitted reservations, review reservation details, approve or decline requests, monitor reservation status
System	Store user accounts and reservation records in MySQL and maintain basic reservation information.

3.2 Project Limitations

· The system will depend on the available computer and local database/network setup.

· Direct integration with online payment gateways, such as GCash and Maya, is not included in the initial project scope.

· The QR code displayed in the system is for display purposes only and is not functional for direct payment processing.

· The system does not have a built-in digital wallet or refund management feature. Refunds cannot be processed directly through the system.

· Real time internet based notifications are outside the initial project scope unless added later.

· Advanced accounting, payroll, and business inventory functions are not included.

· The system will not include a forgot password or password recovery feature in its initial version.

· The initial price computations may not always be accurate and may require further testing, adjustments, and improvement.

· The final system will only include features that can be completed within the project's academic schedule and available resources.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The project is technically feasible for a student development team because it can be created using Java GUI for the desktop application, NetBeans IDE as the development environment for writing, debugging, and running the Java code, MySQL as the database management system, and phpMyAdmin for creating, viewing, and maintaining the database during development. These tools can run on a standard development computer and are appropriate for building the core reservation, transaction, and record management functions required by the project.

4.2 Operational Feasibility

The system is operationally feasible if it follows the printing center's actual reservation process and remains simple enough for students and printing staff to learn. The interface should use clear labels, readable buttons, organized forms, and reservation status information so users can complete the process with minimal confusion.

4.3 Economic Feasibility

The project can be economically feasible because the student version may use free or educational development tools. Possible expenses may include transportation for client meetings, printing and documentation, backup storage, hardware adjustments, and other deployment-related costs. The estimated costs should be compared with the expected benefits of reducing manual work and improving reservation organization.

4.4 Project Charter Summary

Charter Item	Sample Entry
Project Title	Hardbound Printing Reservation & Monitoring System for Four Angel's Copy Center
Project Purpose	Improve the printing reservation and claiming process through a desktop software application.
Primary Users	Students and authorized printing staff / administrators.
Major Deliverable	Working desktop printing reservation system with documentation and basic user guide.
Main Constraint	Limited development time, available hardware, and student-level resources.
Project Success	Core reservation functions work correctly and the client can perform the agreed reservation-management process using the system.

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
Copy Center Owner / Manager	Approves the project direction and provides business requirements.
Printing Staff / Admin	Main operational user who reviews and manages student reservations.
Students/Clients	Primary users who create accounts, submit printing requests, and check reservation status.
Development Team	Analyzes, designs, develops, tests, and documents the proposed system.

5.2 Requirements Gathering Methods

The development team may use the following methods to collect project requirements:

- **Client interview** – To understand the current printing reservation and claiming process.
- **Observation** – To observe how students submit requests and how staff manage printing reservations.
- **Review of Existing Records** – To review details, records, or forms used by the printing center.
- **Follow-up consultation** – To confirm gathered requirements and clarify the proposed system features.

5.3 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow a student to register for an account.
FR-02	The system shall allow registered students to log in.
FR-03	The system shall allow students to enter file and printing details.
FR-04	The system shall allow students to select a claiming date and optional time.
FR-05	The system shall allow students to submit a printing reservation.
FR-06	The system shall save submitted reservation details in the database.
FR-07	The system shall allow students to view the status of their reservations.
FR-08	The system shall allow authorized admins to log in.
FR-09	The system shall allow admins to view submitted reservations.
FR-10	The system shall allow admins to approve or decline reservations.
FR-11	The system shall allow authorized users to review reservation records.
FR-12	The system shall provide navigation for returning to appropriate menus or logging out.

5.4 Non-Functional Requirements

Category	Sample Requirement
Usability	The reservation interface should be simple enough for students and trained staff to use with minimal steps.
Performance	Common actions such as opening forms, saving reservations, and viewing records should respond within a reasonable time on the target computer.
Security	Administrative functions should only be available to authorized users
Reliability	Submitted reservation records should be saved correctly and remain available after restarting the application.
Maintainability	Reservation information and system records should be manageable through the database without rewriting the entire application.

5.5 User Stories and Acceptance Criteria

User Story	Acceptance Criteria
As a student, I want to submit my printing details so that I can reserve a printing request without relying on handwritten forms.	Required printing fields can be entered, the request can be submitted, and the reservation appears with a status
As a student, I want to select a claiming schedule so that I know when I can claim my finished project.	A claiming date can be selected and an optional time can be entered.
As an admin, I want to review reservations so that I can manage printing requests.	Authorized admin can open a list of reservations and view the submitted details.
As an admin, I want to approve or decline requests so that reservation status can be managed.	As an admin, I want to approve or decline requests so that reservation status can be managed

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

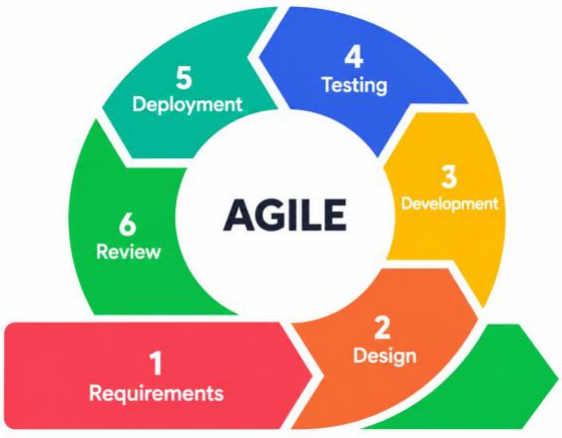
6.1 Development Methodology Used

The project will use an Agile development methodology supported by the Scrum framework. The work will be divided into short development cycles or sprints. During each sprint, the team will select priority tasks from the product backlog, develop and test the selected features, review the completed work, and use feedback to improve the next sprint. This approach helps the team manage changes, track progress, and keep the project focused on the printing center's most important requirements.

6.2 SDLC Model Used

The selected SDLC model for the project is Agile SDLC. Instead of completing the entire system in one long development cycle, the project will be built gradually through repeated iterations. Each iteration may include planning, requirements analysis, design, development, testing, and review. Feedback from the client or instructor can then be considered before the next iteration begins.

6.3 SDLC Model Figure



Figurev1. Agile SDLC Model Used for the Project

6.4 Software and Development Tools Used

The following software and technologies are planned for the development of the printing reservation system:

Software / Technology	Use in the Project
Java GUI	Used to create the application interface and implement the main reservation, user interaction, and administration functions.
NetBeans IDE	Used as the integrated development environment (IDE) for writing, editing, debugging, and running the Java code for the Printing Reservation and Monitoring System.
MySQL	Used as the database management system for user accounts, reservation details, schedules, and reservation status.
phpMyAdmin	Used by the development team to create, view, organize, and maintain the MySQL database during development and maintenance.
Visual Studio Code	Used for debugging, editing, and reviewing project code to identify and fix errors during development.

Java GUI will serve as the application layer that staff interact with, while NetBeans IDE will be used to develop, edit, debug, and run the Java application. MySQL will store the project data, while phpMyAdmin will mainly support database setup and administration during development and maintenance. Visual Studio Code will also be used to edit, review, and debug code to identify and fix errors during development.

6.5 Reason for Selecting the Methodology and SDLC Model

Agile SDLC with Scrum is appropriate for this project because restaurant requirements may become clearer after users see and try the actual ordering screens. Short sprints allow the team to develop important features first, demonstrate progress, receive feedback, and correct problems before moving to the next set of tasks. This approach also supports the project's limited academic schedule by keeping work organized and visible.

6.6 Expected Development Approach

· Prepare and prioritize the initial system requirements.

· Divide the work into small reservation-system features or backlog items.

· Develop a working set of features during each sprint.

· Test completed features before adding more work.

· Present progress to the client or instructor for feedback.

· Improve the system based on accepted feedback.

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stage	Application to the Ordering System
Planning	Identify the printing reservation problem, project objectives, users, scope, available resources, and major constraints.
Requirements Analysis	Interview and observe printing-center users, list functional and non-functional requirements, and confirm reservation rules.
System Design	Prepare screen layouts, navigation flow, database structure, reservation process, and user access design.
Development	Build registration, login, reservation forms, database connection, reservation status, admin management, and record viewing functions.
Testing	Check each function, validate required inputs, test invalid entries, test database operations, and allow users to try the reservation workflow.
Deployment	Install the approved version on the target computer and prepare basic user instructions.
Maintenance	Correct discovered errors, maintain reservation information, back up data, and plan approved future improvements.

8. PROJECT RISK MANAGEMENT

8.1 Possible Risks Discovered

The team should identify events that may affect the project's schedule, cost, scope, or quality. Risks should be realistic and connected to the selected project.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
Client changes important requirements during development.	Medium	High	Confirm requirements per sprint and record approved changes before development continues.
Development takes longer than planned.	Medium	High	Prioritize core reservation features, monitor progress, and adjust low-priority tasks.
Project files or database are lost.	Low	High	Use source control where available and maintain regular local/cloud backups.
A team member becomes unavailable.	Medium	Medium	Document work, share project access, and assign backup responsibilities.
Target computer has limited hardware capability.	Medium	Medium	Test the application on hardware close to the client's actual computer before deployment.
Users find the ordering flow confusing.	Medium	Medium	Perform early user testing and simplify screens based on feedback.
Database connection or saving errors occur.	Medium	High	Test database operations regularly and maintain database backups during development.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The project schedule should show the order of major tasks, expected duration, dependencies, and important milestones.

9.2 Project Schedule

Activity	Duration	Target Period	Main Output
Planning and Client Study	2 days	Week 1	Problem, objectives, scope
Requirements Gathering	2 days	Week 1	Confirmed system requirements
System and UI Design	3 days	Week 1	Interface layouts and database design
Sprint 1 Development	5 days	Week 2	Login, user interface, reservation form
Sprint 2 Development	5 days	Week 3	Reservation management, status monitoring, and claiming schedules
Sprint 3 Development	3 days	Week 4	Database integration, reservation records, and basic computations
Testing and Revision	2 days	Week 4	Tested system and corrected errors
Deployment and Final Documentation	3 days	Week 4	Installed system and final documents

9.3 Project Milestones

Milestone	Target
Requirements approved	End of Week 1
Initial design approved	End of Week 1
Core ordering functions working	End of Week 2
Feature-complete version	End of Week 3
Testing completed	End of Week 4
Final deployment / presentation	End of Week 4

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

The budget should show the resources that may require money during development and deployment. Students should use realistic estimates and identify free resources when applicable.

Cost Item	Basis	Estimated Cost
Development Software	Free / educational tool	PHP 0
Database Software	Free / open-source option	PHP 0
Transportation	Client meetings and testing visits	PHP 300
Printing / Documentation	Drafts, final copies, forms	PHP 100
Backup Storage / USB	Optional project backup	PHP 250
Hardware Adjustment	Allowance for minor peripherals or replacement needs	PHP 300
Contingency	Allowance for unexpected project expenses	PHP 250
TOTAL ESTIMATED COST		PHP 1, 200

10.2 Cost Justification

The sample budget focuses on development and deployment expenses rather than charging for student labor. Transportation and documentation support client meetings and project submission. Backup storage helps protect project files, while the hardware and contingency amounts provide an allowance for possible deployment needs. The actual group should revise each amount based on the tools, equipment, and client conditions of their project.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	Sample Responsibility
Product Owner	Represents the client needs, helps prioritize features, and confirms whether completed work meets agreed requirements.

Scrum Master	Organizes Scrum activities, helps remove project blockers, and keeps the team focused on the sprint goal.
Development Team	Designs, codes, tests, documents, and improves the ordering system.

11.2 Sample Product Backlog

Priority	Backlog Item	Expected Result
High	User registration and login	Students and authorized admins can access the appropriate part of the system.
High	Reservation form	Students can enter complete printing and claiming details.
High	Save reservation	Submitted reservation details are stored in MySQL.
High	Reservation status	Students can view pending, approved, or declined status.
High	Admin reservation view	Admin can review submitted reservations.
High	Approve / decline reservation	Admin can update the status of a request.
Medium	Reservation history	Authorized users can review saved reservation records.
Medium	File upload support	Students can attach a file if included in the final implementation.
Low	Additional reports	More detailed reports may be added if time allows.

11.3 Sample Sprint Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Create the basic application interface.	Registration, login, navigation, and basic reservation form.
Sprint 2	Complete the core reservation process.	Printing details, claiming schedule, validation, database saving, and reservation status.
Sprint 3	Add administration and monitoring functions.	Admin reservation view, approval/decline functions, and system improvements.

Sprint 4	Prepare the system for final use.	Testing, bug fixes, user feedback, database checking, and deployment preparation.

11.4 Scrum Events

Scrum Event	How the Team Will Apply It
Sprint Planning	Select backlog items that can realistically be completed during the sprint and agree on the sprint goal.
Daily Scrum / Team Check-in	Briefly discuss completed work, current tasks, and blockers.
Sprint Review	Demonstrate completed reservation functions to the client, instructor, or project group and gather feedback.
Sprint Retrospective	Discuss what worked, what caused problems, and what the team should improve in the next sprint.

11.5 Simple Scrum Progress Tracking

The team may use a simple task board with three basic columns: To Do, In Progress, and Done. Each backlog item should have an assigned member and status. At the end of every sprint, the team should compare planned work with completed work and use the result to improve the next sprint.

12. PROJECT SUMMARY

The proposed the proposed Hardbound Printing Reservation and Monitoring System is intended to improve the printing reservation process of Four Angel's Copy Center by providing a more organized method of submitting printing requests, recording reservation details, managing claiming schedules, and reviewing reservation status. The project focuses on the functions needed by students and printing staff, which helps keep the scope realistic for a student development team.

The project can be developed using commonly available software and hardware if the final requirements are properly confirmed with the client. Using Agile and Scrum allows the team to develop the system in manageable parts, test features early, and use feedback to improve the final product. The risk register, schedule, and budget provide additional controls so that the team can manage the project more systematically.